idea with a lower-dimensional analogy involving twodimensional spheres. We see that a two-dimensional membrane can smear itself around a two-dimensional sphere (which itself is sitting inside a Calabi-Yau space positioned at some location in the extended dimensions). Someone looking through the extended dimensions toward this location will sense the wrapped brane by its mass and the force charges it carries, properties that Horowitz and Strominger had shown would look just like those of a black hole. Moreover, in Strominger's 1995 breakthrough paper, he argued that the mass of the three-brane—the mass of the black hole, that is—is proportional to the volume of the three-dimensional sphere it wraps: The bigger the volume of the sphere, the bigger the three-brane must be in order to wrap around it, and the more massive it becomes. Similarly, the smaller the volume of the sphere, the smaller the mass of the three-brane that wraps it. As this sphere collapses, then, a three-brane that wraps around the sphere, which is perceived as a black hole, appears to become ever lighter. When the three-dimensional sphere has collapsed to a pinched point, the corresponding black hole—brace yourself—is massless. Although it sounds completely mysterious—what in the world is a massless black hole?—we will soon connect this enigma with more familiar string physics.

The second ingredient we need to recall is that the number of holes in a Calabi-Yau shape, as discussed in Chapter 9, determines the number of low-energy, and hence low-mass, vibrational string patterns, the patterns that can possibly account for the particles in Table 1.1 as well as the force carriers. Since the space-tearing conifold transitions change the number of holes (as, for example in Figure 13.3, in which the hole of the doughnut is eliminated by the tearing/repairing process), we expect a change in the number of low-mass vibrational patterns. Indeed, when Morrison, Strominger, and I studied this in detail, we found that as a new two-dimensional sphere replaces the pinched three-dimensional sphere in the curled-up Calabi-Yau dimensions, the number of massless string vibrational patterns increases by exactly one. (The example of the doughnut turning into a beach ball in Figure 13.3 would lead you to believe that the number of holes—and thus the number of patterns—decreases, but this proves to be a misleading property of the lower-dimensional analogy.)